# IF-CPD Experiments on the Supplementary Dataset

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In this PDF, we apply the IF-CPD method to three publicly available real-world datasets: the Enron Email Network Data [Coh15, DM20], the MIT Reality Mining Proximity Networks [EP06, DZ23], and the U.S. electricity generation dataset [DZ23].We have presented the experimental results in visual form and provided a brief explanation of the findings. **These experimental conclusions demonstrate the effectiveness and reliability of our method.**

Table 1: Some of the indicators shown in the figure.

| **Indicator** | **Definition** |
|---|---|
| Total Emails per Week | The total number of emails sent and received by all users during the week. This is a metric used to measure the overall activity of the network. |
| Network Density | Network density = actual number of edges / maximum possible number of edges. A high density indicates that the connections between nodes are tighter. |
| Mean Edge Weight (non-zero) | The average weight of non-zero edges. The weight is the number of emails and reflects the intensity of interaction. |
| Max Edge Weight | The maximum edge weight between any two nodes during that week represents the strongest connection. |
| Active Nodes | The number of nodes that have participated in at least one interaction reflects the size of the network. |
| Reciprocity | Reciprocity = Number of mutual edges / Total number of edges. High reciprocity indicates that interactions are more symmetrical. |

# Experiments on the Enron Email Network Data

Figure 1: The experimental period spans 183 weeks, from November 1998 to June 2002, with the central segment shown in the figure. The detected change point occurs at week 87, closely aligning with the week 88 reported in [DM20]. This timing falls precisely before a pivotal event for Enron—August 23, 2000, when the company's stock price reached an all-time high.

# Experiments on the MIT Reality Mining Proximity Networks

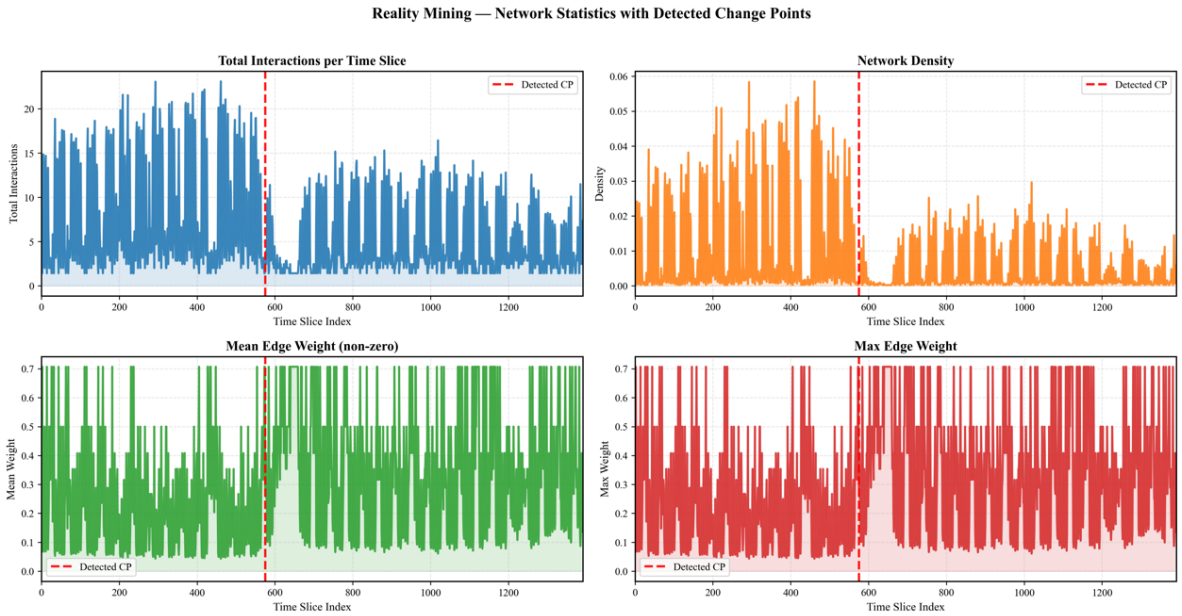


Figure 2: The detected change point corresponds to the 575th time point. Given a temporal resolution of four hours per interval, this change point falls on December 18, 2004—a date that closely aligns with December 15, 2004, as estimated in [DZ23]. This timing lies within the final examination period.

# Experiments on the U.S. electricity generation dataset

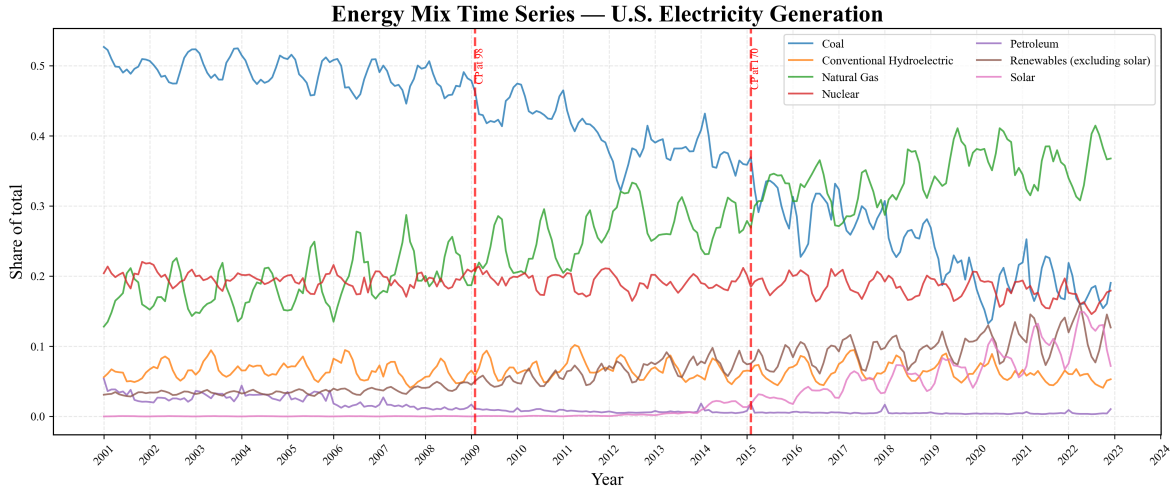


Figure 3: Two change points are detected, one in 2009 and the other in 2015. The latter aligns with the finding reported in [DZ23]. The 2009 change point coincides with the passage of the American Clean Energy and Security Act in the United States. Meanwhile, the availability of cheap natural gas from the shale gas revolution triggered a long-term decline in coal-fired power generation during the economic crisis.

## References


[Coh15] William W. Cohen. Enron email dataset. https://www.cs.cmu.edu/~enron/, 2015. Maintained by William W. Cohen, Machine Learning Department, Carnegie Mellon University.

[DM20] Paromita Dubey and Hans-Georg Müller. Fréchet change-point detection. *The Annals of Statistics*, 48(6):3312–3335, 2020.

[DZ23] Paromita Dubey and Minxing Zheng. Change point detection for random objects using distance profiles. *arXiv preprint arXiv:2311.16025*, 2023.

[EP06] Nathan Eagle and Alex (Sandy) Pentland. Reality mining: sensing complex social systems. *Personal and Ubiquitous Computing*, 10(4):255–268, March 2006.